PAC autotuning comparison | no payload | ϕ_d = 1 rad | 10 s/circle

12 circles; errors = desired - measured; invalid samples retained as gaps

Desired — Candidate 1 (baseline) — Candidate 5 — Candidate 8 — Candidate 9 (smooth)

Azimuth tracking

θ, unwrapped (rad)

Bend-angle tracking

ϕ (rad)

Azimuth error

e_θ (rad)

Bend-angle error

e_ϕ (rad)

Length error: l_2

e_{l_2} (mm)

Time since circle start (s)

Length error: l_3

e_{l_3} (mm)

Time since circle start (s)